STRATHMORE UNIVERSITY

Strathmore Institute of Mathematical Sciences

MSc in Data Science and Analytics

CONCEPT NOTE

Sh73 Billion Collected, 1,795 Homes Built: Chaining Vulnerability, Efficiency, and Allocation in Kenya's Affordable Housing Programme

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1. Background and Context

Kenya's Affordable Housing Programme (AHP), established under the Affordable Housing Act, 2024 and funded through the statutory Housing Levy, targets 200,000 housing units per year across the country's 47 devolved counties. In FY2024/25 the Kenya Revenue Authority collected KSh 73.2 billion in Levy receipts, 115.8% of the National Treasury's own KSh 63.2 billion target, with cumulative collections since 2024 now past KSh 170 billion. Yet only 1,795 units were completed nationally in the same period, and a Treasury report tabled in Parliament confirmed that more than KSh 30 billion of already collected proceeds remained undisbursed (Business Daily Africa, 2025). County-level site selection and budget allocation under the AHP currently rest on manual, cost-centric judgement rather than on a systematic, data-driven assessment of which households are most financially vulnerable, which counties can deliver housing most efficiently, and how a constrained annual budget should be rationed between them.

This proposal draws its structural justification from Kenya's constitutional seam for devolved housing functions (the Fourth Schedule of the 2010 Constitution, read with Article 43(1)(b)) and from the AHP's standing as a flagship pillar of Kenya Vision 2030's Bottom-Up Economic Transformation Agenda, and positions the Housing Financial Vulnerability Score (HFVS) framework as a systematic, evidence-based replacement for the AHP's current manual site-selection process.

2. Problem Statement

The Levy's revenue side is working roughly as designed; its delivery side is not. County-level allocation of the Levy fund and unit quota is decided through fragmented, manual, tabular reviews that cannot distinguish a county's genuine housing-delivery capacity from its revenue entitlement, its population share, or its historical precedent, and that offer no mechanism for tracing a shilling of allocated funding back to the household-level risk evidence that should justify it. Without an automated, data-driven decision-support pipeline, state housing investment risks exacerbating spatial segregation, financial default among beneficiary households, and inefficient capital deployment across devolved units. The central planning question this raises is not whether money exists, but where, precisely, the constraint actually binds: on the budget, or on the physical capacity to build, permit, and hand over completed units. Current allocation practice has no systematic way to answer that question for any individual county, let alone all 47 simultaneously.

3. Purpose and Objectives

The overall purpose of the study is to design, validate, and deploy a single, chained CRISP-DM decision-support pipeline that operationalises household financial vulnerability, benchmarks county-level delivery efficiency, and optimises Housing Levy allocation under real budget, quota, and capacity constraints. Four research objectives structure the work:

- **RO1.** Develop and validate a leakage-audited, spatially cross-validated machine-learning model of household housing financial vulnerability (HFVS) from the Kenya Housing Survey (KHS) 2023/24 micro-data, built as a five-pillar composite (financial stress, tenure insecurity, physical hazard, structural quality, utility deprivation) with pillar weights derived empirically via SHAP attribution rather than assumed equal, and housing cost burden predicted through a progression of models (gradient boosting, a stacking ensemble, and a two-stage hurdle architecture for the zero-inflated burden distribution).
- **RO2.** Formulate a non-radial, bootstrap-validated Data Envelopment Analysis (Slack-Based Measure) model of county-level housing-delivery efficiency across Kenya's 47 counties, incorporating survey-weighted HFVS aggregates in its input to output specification, with Simar and Wilson bootstrap confidence intervals, so as to operationalise delivery capacity as a measurable, county-specific quantity rather than an assumption.
- **RO3.** Design and solve a Mixed-Integer Linear Programme, with integer delivery units and binary county activation, that allocates the statutory Levy fund and unit quota across counties under an explicit, county-specific delivery-capacity constraint held distinct from the budget constraint, comparing a capital-concentration regime against a universal-coverage regime, in order to establish, county by county, whether revenue or delivery capacity is the constraint actually binding programme impact.
- **RO4.** Validate the chained pipeline's resulting county rankings and constraint diagnoses against an independent qualitative case study, and assess the framework's transferability to other East African Community contexts.

4. Justification and Significance

At the global and continental level this work speaks to SDG 1, SDG 10, and SDG 11 (target 11.1 on adequate, safe, and affordable housing), and to African Union Agenda 2063, Aspiration 1. Its distinguishing empirical contribution is a per-county, per-regime binding-constraint diagnostic: for every county and every allocation regime, the pipeline identifies whether capacity, budget, or the national unit quota is the constraint actually preventing more units from being delivered. This reframes the policy conversation away from a single number (how much money is available) toward a structural diagnosis (what, specifically, is stopping this county from building more), which is directly actionable for the National Treasury, the State Department for Housing, and county governments, and makes the concentration versus coverage trade-off explicit and quantifiable in shillings rather than an implicit policy choice.

5. Methodology Overview

The study follows a CRISP-DM structure across three chained modelling tiers on a single data spine, the Kenya Housing Survey 2023/24 (21,347 households, 47 counties, joined across six thematic layers), with strict leakage prevention, train-fitted transformations, and spatial (county-grouped) cross-validation, followed by an independent validation layer:

- **Tier 1 (Household).** Five-pillar HFVS construction with SHAP-derived pillar weights, followed by housing cost burden prediction (baseline linear regression through a stacking ensemble and a renter-specific two-stage hurdle model), with an explicit equity audit of prediction error by income quartile.
- **Tier 2 (County).** Non-radial SBM data envelopment analysis of county delivery efficiency, benchmarked against radial CCR/BCC comparators, with Simar and Wilson bootstrap confidence intervals around each efficiency score, feeding a county-specific delivery-capacity ceiling forward.
- **Tier 3 (National).** Integer, binary-activation MILP allocating Housing Levy units across counties under sourced budget (KSh 73.2B), quota, and independently derived capacity-ceiling constraints, run under a capital-concentration regime and a universal-coverage regime, with population-weighted vulnerability driving the objective.
- **Validation.** The resulting county rankings and constraint diagnoses are checked against an independent qualitative case study, as the external anchor for Tier 1 to 3's internal validation metrics.

6. Key Preliminary Findings

Household-level results show 47.1% of surveyed households in the High or Critical HFVS tier, against only 4.7% flagged by a raw cost-burden threshold, indicating that a composite, multi-pillar score surfaces materially more at-risk households than a single financial ratio would. County efficiency benchmarking shows a meaningful radial-masking gap between the SBM and CCR/BCC scores, consistent with the literature's own critique of radial DEA at this sample scale. Under the delivery-constrained MILP (a 6,000-unit annual quota, the realistic near-term planning number given demonstrated build rates), the capital-concentration regime activates 26 of 47 counties, allocates the full 6,000-unit quota, and spends approximately KSh 19.75 billion of the KSh 73.2 billion available. Of the 26 activated counties, 24 are constrained by their own delivery-capacity ceiling and 2 by the national quota, with none constrained by budget. Because 6,000 units at the model's average per-unit cost comes to only around 27% of the available Levy budget, budget was always unlikely to bind at this quota level; the more informative result is the capacity and quota split among activated counties, and a sensitivity pass identifying the quota level at which budget would begin to bind is planned as a robustness check. This confirms that, at a realistic delivery-constrained quota, construction and delivery capacity, not levy revenue, is the binding national constraint, a finding that directly targets AHP policy toward capacity-building rather than revenue mobilisation alone.

7. Expected Outputs

- A dissertation manuscript in Elsevier CAS format, with an accompanying methodology, results, and discussion structure mirroring the four research objectives.
- A reproducible analytical pipeline (Jupyter notebook) covering data cleaning, HFVS construction, Tier 1 ML, Tier 2 DEA, and Tier 3 MILP, with documented, sourced assumptions and an explicit limitations register.
- A deployed, browsable dashboard presenting household, county, and national-level results for non-technical policy audiences.
- An Excel-based MILP solver replica, cross-validated against the notebook's own output, allowing a reviewer without programming access to reproduce and audit the Tier 3 allocation.
- An independent qualitative case-study validation of the resulting county rankings and constraint diagnoses (RO4).

8. Scope and Limitations

The analysis is confined to the 47 counties and the household-level fields captured in the Kenya Housing Survey 2023/24, and treats the MILP as a single-year planning model rather than a multi-year rollout schedule. HFVS-derived quantities enter the pipeline at two points, as a Tier 2 DEA output (county financial stability) and again as the vulnerability term in the Tier 3 objective. This is a deliberate design choice to keep efficiency and need as distinct lenses, but one that will be justified explicitly in the dissertation to pre-empt a double-counting critique. Two items are flagged as open before defence: the Simar and Wilson bootstrap confidence intervals for frontier counties require verification, as the printed bounds for those counties currently look mechanically inconsistent with their point estimates; and unit-cost sensitivity has not yet been re-run across the KSh 1.05 million to KSh 1.95 million per-unit band suggested by general 2026 residential construction rate surveys. Both will be resolved ahead of the proposal defence, since they bear directly on the validity of the Tier 2 rankings and the Tier 3 budget-utilisation figures reported above.